

3.5 Detailed specific example: Ecuador



Ecuador is a country in the north-western region of South America. It borders Colombia in the north-east, Peru in the south and south-east, and the Pacific Ocean in the west.

Ecuador is one of the most environmentally diverse countries in the world. It is situated on the equator and includes part of the Andes Mountain range, as well as part of the Amazon basin. It is divided into four main physical regions: the coastal region (Costa), the highlands (Sierra), the eastern region (Oriente), and the Galapagos islands. Most of the country has a humid tropical climate with high rates of rainfall and warm temperatures throughout the year.

The Oriente region has continuous and abundant rainfall (3000 mm to 6000 mm per year) with constantly high daytime temperatures (26 °C to 28 °C) throughout the year. This region is where tropical rainforests thrive. The tropical rainforest has high biodiversity due to the large trees and diverse vegetation.



Figure 3.5: View of the rainforest from an observation tower in Ecuador

Source A: Climate data for the Oriente region

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temperature °C	26	27	27	26	26	25	24	24	25	26	26	27
Rainfall (mm)	407	465	556	439	315	196	164	111	154	374	469	422

TIP

Combined climate graphs show temperature as a line graph and rainfall as a bar graph.

TASK 1: Study Source A

- Using the data in Source A, **draw a climate graph** showing both temperature and rainfall. Put the temperature scale on the right-hand y-axis and the rainfall scale on the left-hand y-axis.
- With reference to the graph you have created**, describe the climate for the Oriente region.
- Explain why the climate in the Oriente region is good for rapid plant growth.

Threats to the tropical rainforest of Ecuador

Deforestation is the greatest threat to the rainforests in Ecuador. There are many reasons for deforestation, outlined in Table 3.5. In the mid-2000s, Ecuador had the highest deforestation rate in South America, putting great pressure on the rainforests.

Table 3.5: Causes of deforestation in the Ecuador rainforest

Oil mining	Expansion of the oil mining industry, including the construction of roads, pipelines, wells, camps/towns, and helipads has resulted in large areas of rainforest being destroyed. More than 6470 km ² of rainforest are reported to have been cleared for mining since the 1970s.
Timber harvesting	In the Guayas River valley, the balsa tree is felled for its light-weight wood. Other hardwood trees are felled for use in construction and furniture.
Agriculture	In the Oriente region, small-scale ranches and subsistence farmers clear the forested land. In addition, the global demand for palm oil has seen areas of the rainforest cleared to grow plantations.
Medicinal	Cinchona trees were an important source of quinine, historically used to treat malaria. The cinchona, which occurs in the Oriente region, is now endangered. In the 21st century, many plants are harvested for their medicinal qualities such as the Curare plant, for modern anaesthetics and to treat muscular disorders.
Urbanisation	Forested areas have been cleared to build towns and roads.



Figure 3.6: Deforestation in the tropical rainforest

Impacts of human activity on the tropical rainforest of Ecuador

Tropical rainforests have complex biodiversity where organisms rely on a balance between biotic interactions and abiotic processes. This means that this ecosystem is vulnerable to change. This balance can be seen in Ecuador. The ongoing deforestation of tropical rainforest areas negatively affects neighbouring locations which have not yet been directly affected by human economic activities. Figure 3.7 outlines some of the impacts of human activity on the rainforest.

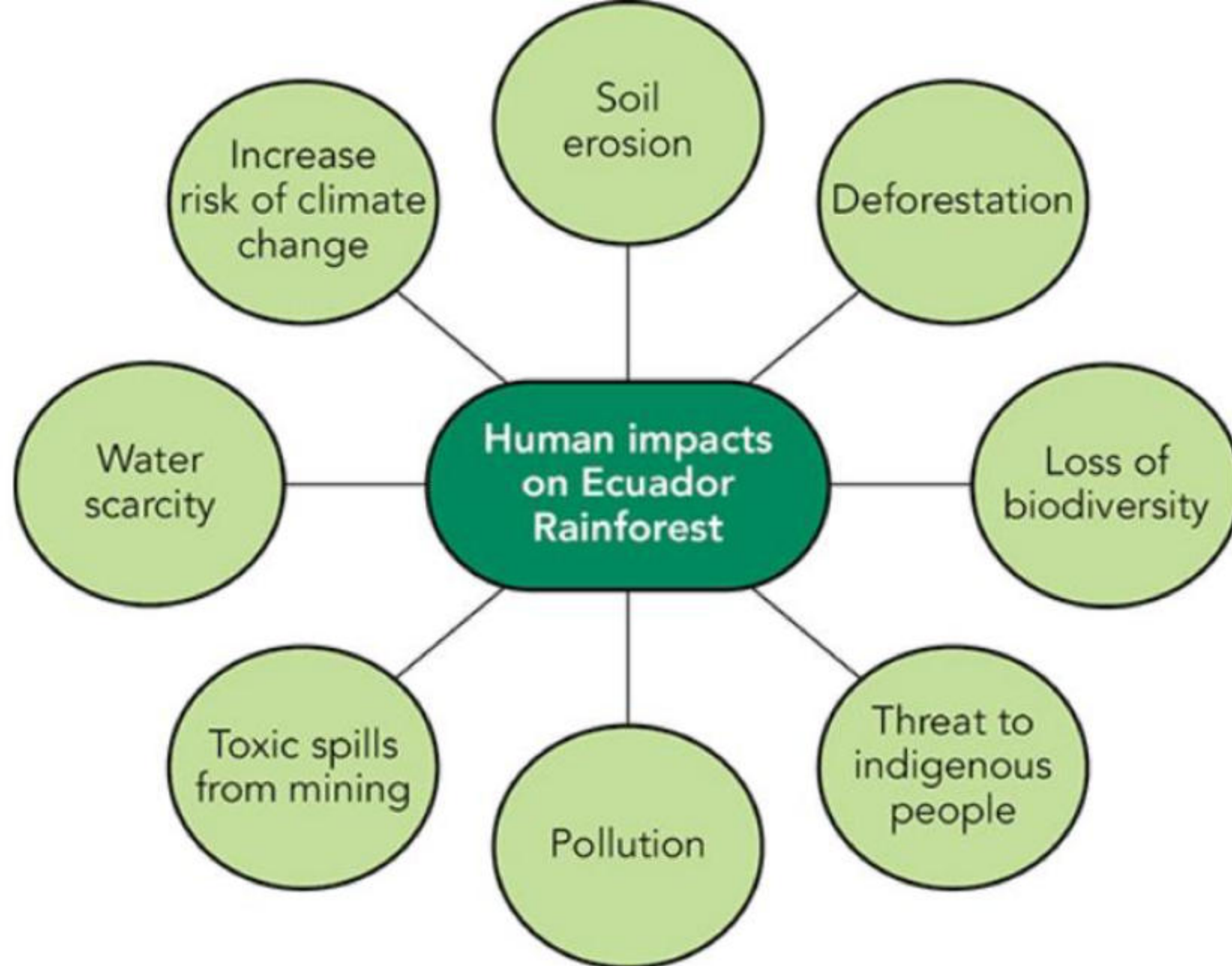
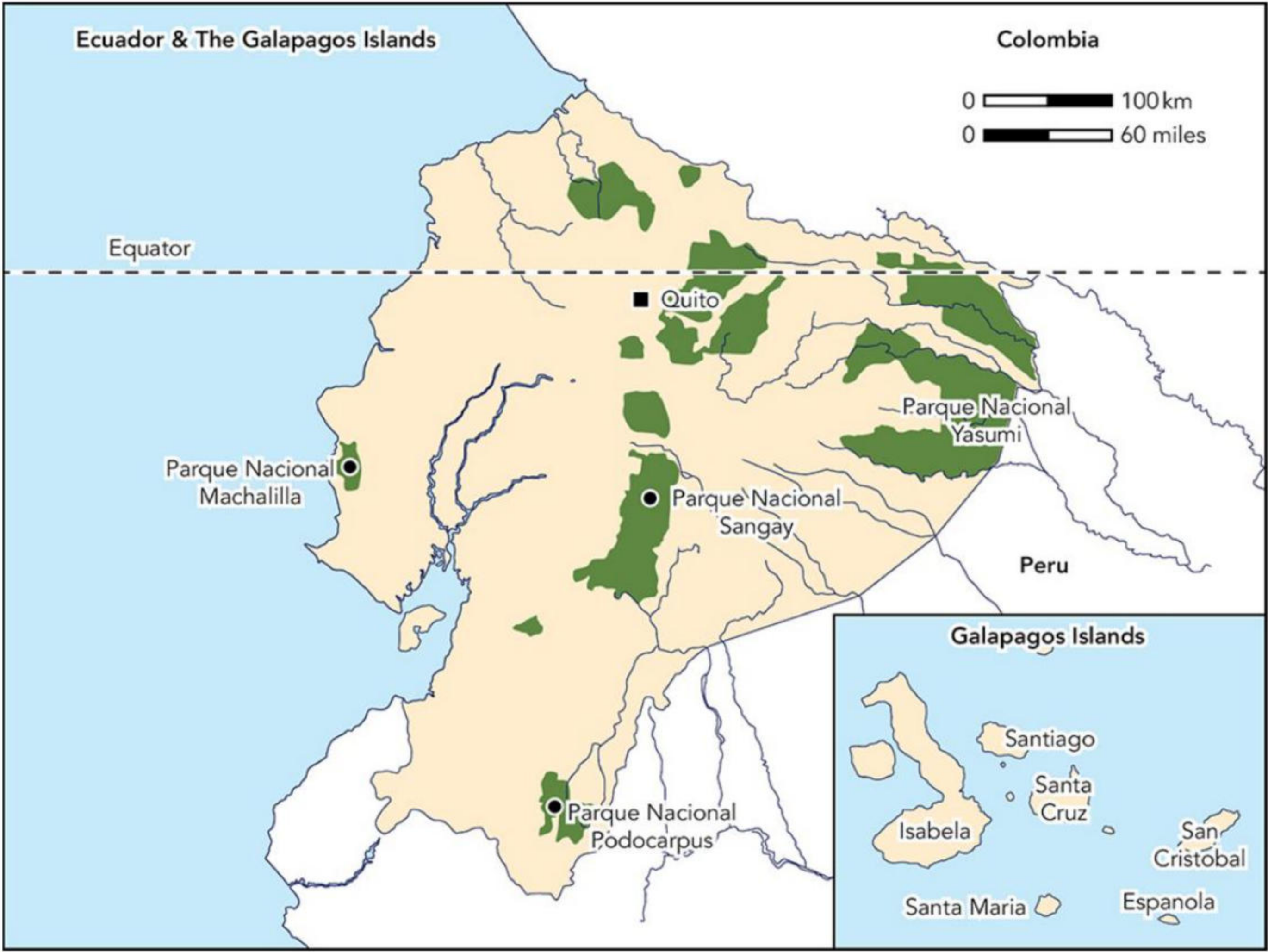


Figure 3.7: Impacts of human activity on the Ecuador rainforest



Source B: A map of Ecuador showing the national parks

Source B shows the national parks in Ecuador. National parks are natural areas which are protected for conservation purposes.

- a Describe the distribution of national parks in Ecuador.
- b Explain the impact of drilling for oil in the national parks on the rainforest.
- c Why do you think national parks are important?

Sustainable management of the tropical rainforest of Ecuador

To manage deforestation and make the use of forested areas more sustainable, Ecuador has implemented a number of successful policies and positive incentives for reducing deforestation and rainforest degradation. These are outlined in Table 3.6.

Table 3.6: Ecuador has put a number of strategies and techniques in place to protect and sustainably manage its rainforests

Policies and action plans	The Ecuador government has devised a number of policies and action plans to protect the rainforest. The Socio Bosque programme began in 2008 and is a scheme for the conservation of rainforests, mangroves, and other native ecosystems in Ecuador. In 2016, Ecuador’s National Assembly passed a bill on land use and land management to further protect the rainforests and biodiversity of Ecuador. To date, these initiatives have reduced the rate of deforestation by nearly 50% and include: 1 The development of land-use management policies to protect rainforests. This limits land use in areas of rainforest identified for conservation. 2 The transition to sustainable development, with the government considering the best ways of protecting rainforests for future generations. This includes conservation and land use management of forested areas. 3 Rainforest management to increase protection of rainforests. Rainforest management includes the allocation of permits for harvesting wood and medicinal plants, protection of undisturbed rainforest, and rehabilitation of areas that have been damaged. 4 Reforestation of key rainforest areas. For example, the World Land Trust planted 30 000 trees to help reforest an area of the Buenaventura Reserve. 5 Identify and prioritise policies and measures to address the causes of deforestation. For example, paying landowners to conserve the rainforest rather than clearing them.
Amazonian community	Including indigenous Amazonian communities in the maintenance of 1590 km² of rainforest to promote rainforest conservation and improve the quality of life for the communities. This is an initiative of the Ministry of Environment, Water and Ecological Transition, which provides direct transfer of resources to indigenous communities, such as the Kichwa, Shiwiar, and Tsáchila nations. This promotes rainforest conservation and improving the quality of life of the communities.
Rainforest-use permits	The implementation of permits for people obtaining resources (such as medicinal plants, food sources, or timber) from the rainforest, with fines for those breaking the law.
Sustainable farming	Introducing sustainable farming methods for crops like coffee, cocoa, oil palm, and livestock, to improve the protection of rainforests and income of farmers. These ‘deforestation-free’ products serve a global market that values environmentally conscious products.
Rainforest reserves	Ecuador has protected parts of the rainforest by establishing rainforest reserves. These reserves are working to safeguard rainforests and develop protected wildlife corridors between isolated patches of rainforest.
Biosphere reserves	Ecuador has established several internationally recognised biosphere reserves. They are designed to protect rainforest while allowing sustainable use of the forested areas. The core areas are protected from human activity, while the buffer and transition zones allow varying levels of human access. Being internationally recognised means that they function within very strict and monitored environmental regulations.

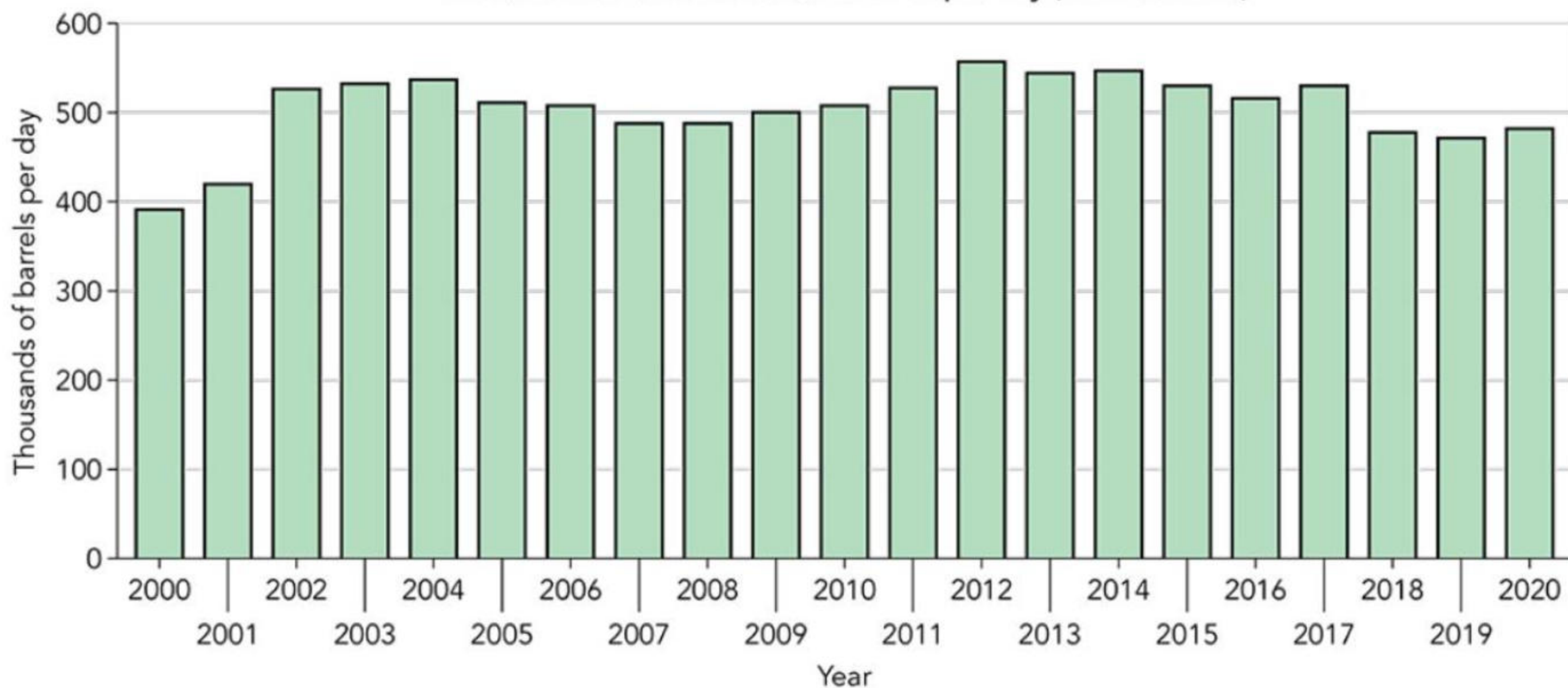
Research has found that over 160 different plant species are sold in the traditional markets in Ecuador. The uses of traditional medicinal plants in Ecuador are related to a long history of the people living in or off the rainforest ecosystem. The plants sold are used to treat illnesses related to digestive, respiratory, skin, and sensory ailments.

Source : Medicinal plants research

TASK 3: Study Source

- a Using information from Source C, explain why the uncontrolled harvesting of medicinal plants is a threat to the rainforest. Consider the ongoing increase in population size.
- b Explain how the Ecuadorian government could manage medicinal plant harvesting to protect the rainforests.

Production in thousands barrels of oil per day (2000 to 2020)



Source **D**: Oil production in Ecuador (2000 to 2020)

► Image Description

TASK 4: Study Source **D**

- Which year had the highest production of oil?
- Describe the trend** in oil production between 2000 and 2020.
- Do you think the oil production rates suggest that policies have been put in place between 2000 and 2020 to protect the rainforests from oil mining? Explain your answer.

133

PROJECT

In pairs or small groups, carry out research about ecotourism to either a tropical rainforest area or a polar region.

- identify the location/s that tourists would be travelling to
- identify the attractions for tourists to the area
- identify the types of accommodation and transport that would be used
- include climate or information on seasons if relevant
- consider the type of tourist that you want to attract to the area
- include how the travel is sustainable and protects the environment being visited.

Create an ecotourist travel brochure that informs the reader about the area and why they would want to travel there.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics.

This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
give the main characteristics of the Antarctic and tropical rainforest ecosystems			
explain the biotic and abiotic factors influencing the characteristics of the Antarctic and tropical rainforest ecosystems			
outline the uses of and threats to the resources from the Antarctic and tropical rainforest ecosystems			
explain the impacts of the destruction of the Antarctic and tropical rainforest ecosystems			
describe and evaluate the sustainable management of the Antarctic and tropical rainforest ecosystems.			

> Topic 4

Tectonic hazards

LEARNING INTENTIONS

By the end of this topic you will be able to:

- describe the characteristics of the Earth's layers, tectonic plates, volcanoes, and earthquakes
- explain the main hazards of earthquakes and volcanoes
- give reasons for why people live in earthquake zones
- explain the causes and characteristics of earthquakes
- give the impacts of earthquakes and volcanoes on people, the economy, and the environment
- explain strategies for managing the risks from earthquakes and volcanoes.

GETTING STARTED

Working in pairs or small groups, discuss the following:

- 1 Name one recent volcanic eruption or earthquake that you have heard of in the news.
- 2 Give reasons for why you think that earthquake or volcanic eruption occurred.
- 3 Name two hazards that can cause damage from either an earthquake or a volcanic eruption, such as injury or building collapse.
- 4 Discuss what you think can be done to make people who live in areas with volcanic activity safe from natural disasters like a volcanic eruption.